

HEP Agent Simulation Framework

Model Documentation

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Contents

1	Introduction	2
2	Environment Model	3
2.1	Grid Structure	3
2.2	Habitat Suitability (HEP)	3
3	Agent Dynamics	4
3.1	Population Structure	4
3.2	Movement Rules	4
3.3	Demographics	4
3.3.1	Birth	4
3.3.2	Death	4
4	Interaction Model	5
4.1	Resource Competition	5

Chapter 1

Introduction

This document describes the mathematical and logical model underlying the Human Ecology Project (HEP) Agent Simulation.

Chapter 2

Environment Model

2.1 Grid Structure

Description of the spatial grid, resolution, and coordinate system.

2.2 Habitat Suitability (HEP)

Description of the HEP index and how it influences agent survival and movement.

Chapter 3

Agent Dynamics

3.1 Population Structure

Agents are defined by a set of intrinsic and extrinsic attributes:

- **Age**: Tracked in ticks (1 tick $\approx$ 1 week).
- **Gender**: Binary classification (Male/Female).
- **Resources**: Energy levels accumulated from grid cells.

3.2 Movement Rules

Logic governing agent migration and dispersal.

3.3 Demographics

The demographics model manages population growth and mortality through a combination of individual agent rules and macroscopic feedback loops.

3.3.1 Birth Model (Verhulst Pressure)

To ensure population stability and prevent exponential runaway, the simulation utilizes a **macroscopic fertility scaling factor** ($K_{fertility}$).

The target birth rate ϕ_{target} is calculated using the Verhulst logistic equation:

$$\phi_{target} = r \cdot \left(1 - \frac{N}{N_C}\right) \quad (3.1)$$

Where:

- r : Intrinsic growth rate.
- N : Current total population count.
- N_C : Carrying capacity of the environment.

The actual fertility of each female agent is then scaled by $K_{fertility}$ to match this target, bounded by K_{min} and K_{max} to prevent demographic collapse or excessive volatility.

3.3.2 Death Model

Mortality is driven by multiple factors:

- **Natural Mortality:** Age-dependent risk.
- **Starvation:** Triggered when an agent's resources fall below a critical threshold.
- **Environmental Risk:** Probability of death based on HEP suitability (e.g., drowning in water cells).

Chapter 4

Interaction Model

4.1 Resource Competition

How agents compete for limited resources in a grid cell.